



St Catherine's
School
Waverley, Sydney

Student Number:.....

Year 12

Extension 1 mathematics-Trial HSC examination

3 August 2011

Time allowed: 2 hours

Reading time: 5 minutes

Course weighting: 40%

General Instructions

- Attempt ALL questions
- Write your Student NUMBER on every booklet used.
- Write using blue or black pen
- Board approved calculators and stencils can be used
- A table of standard integrals is provided
- Show necessary working
- Organise Q 1 to 4 in one bundle and Q 4 to 7 in another

Attempt Questions 1 to 7-All questions are of equal value

Total marks /84

Question 1

- a) Use the result $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ to find the value of $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ 2m
- b) (i) Show that the curves $y = \log_e x$ and $xy = e$ meet at the point $(e, 1)$. 1m
- (ii) If α is the acute angle between the tangents at the point of intersection show that $\tan \alpha = \frac{2e}{e^2 - 1}$ 3m
- c) Eight people are seated around a table.
- (i) How many arrangements are possible if there are no restrictions? 1m
- (ii) Laura, Sally and Lauren want to sit together. How many arrangements are possible? 2m
- d) Integrate using the substitution $u = 1 - x$ $\int \frac{x}{\sqrt{1-x}} dx$ 3m

Question 2

Start a new page

- a) Evaluate and leave in exact form.

$$\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{9-2x^2}}$$

3m

- b) $P(x)$ is an even polynomial of degree 4 i.e. $P(-x) = P(x)$.

Two of its zeros are at $x = 1$ and $x = 2$

- (i) Find the other two roots.

2m

- (ii) If $P(0)=8$, Write $P(x)$ in factored form.

2m

- c) (i) Show that $\sqrt{3} \cos x - \sin x = 2 \cos(x + \frac{\pi}{6})$

1m

- (ii) Sketch the function $y = \sqrt{3} \cos x - \sin x$ in the domain $0 \leq x \leq 2\pi$ clearly labelling the end points and the turning points.

2m

- (iii) Find the general solution to the equation $\sqrt{3} \cos x - \sin x = 1$

2m

Question 3

Start a new page

- a) (i) Write the expansion of $(5 + 2x)^{15}$ in ascending powers of x and show that

$$\frac{\text{coefficient of } t_{r+1}}{\text{coefficient of } t_r} = \frac{32-2r}{5r}$$

3m

where t_r is the r^{th} term in the above expansion.

- (ii) Hence find the greatest coefficient

2m

(You may leave the answer in terms of the binomial coefficient.)

- b) Find the constant term in the expansion of

3m

$$\left(x - \frac{1}{2x^3}\right)^{12}$$

You may leave the answer in terms of the binomial coefficient.

- c) A machine is known to produce items of which 3% are too short, 7% are too long and 90% are satisfactory. A random sample of twelve items is taken from the sample.

Find the probability (correct to two decimal places) that

- (i) At most one of these items is too long

2m

- (ii) At least ten of these items are satisfactory.

2m

Question 4

Start a new page

- a) Consider the function

$$f(x) = \frac{e^x}{x-1}$$

- (i) Show that $(2, e^2)$ is a minimum turning point. 3m

- (ii) Determine all the vertical and horizontal asymptotes of the curve $y=f(x)$ and sketch the graph $y=f(x)$ including any intercepts with the coordinate axes. 3m

- b) Let $f(x) = 4x - x^2$ for $x \geq 2$.

- (i) Sketch the function $y = f(x)$ and $y = f^{-1}(x)$ on the same set of axes. 2m

- (ii) Find an expression for $y = f^{-1}(x)$ 2m

- (iii) Find the point of intersection of $y = f(x)$ and $y = f^{-1}(x)$ 2m

Question 5

START A NEW BOOKLET

- a) Consider the parabola $x^2 = 4ay$. Let $S(0, a)$ be the focus.

The tangents at points $P: (2ap, ap^2)$ and $Q: (2aq, aq^2)$ meet at the point T .

- (i) Show that the coordinates of T is $(a(p+q), apq)$

You may assume that the equation of the tangent at any point 2m

$P: (2ap, ap^2)$ is $y = px - ap^2$

- (ii) Show that $SP = a(p^2 + 1)$ (Draw a figure) 1m

- (iii) P and Q move on the parabola so that $SP + SQ = 4a$

Find the locus of T . 2m

- b) A particle is moving in a straight line in Simple Harmonic Motion 3m
according to the equation

$$\frac{d^2y}{dx^2} = -x$$

Initially the particle is at $x = 1$ and has a velocity of 1cm/sec .

Show that $x = \sqrt{2} \cos(t - \frac{\pi}{4})$

- c) Suppose that the cubic function $f(x) = x^3 + ax^2 + bx + c$ has a relative maximum at $x = \alpha$ and a relative minimum at $x = \beta$.

- (i) Prove that $\alpha + \beta = \frac{-2a}{3}$ 2m

- (ii) Deduce that the point of inflexion occurs at

$$x = \frac{\alpha + \beta}{2}$$

2m

Question 6

Start a new page

a) Let $f(x) = \frac{1}{2} \cos^{-1}(1 - 3x)$

(i) Find the domain and the range

(ii) Sketch the function clearly

b) Integrate using the substitution $x = 3 \sin \theta$

$$\int_0^3 \frac{x^2 dx}{\sqrt{9 - x^2}}$$

c) If $f(x) = \cos^{-1}(\sin x)$

(i) Show that $f'(x) = \pm 1$. (note that $\sqrt{x^2} = |x|$)
State the values of x for which it is 1 or -1 in the domain
 $-\pi \leq x \leq \pi$

(ii) Hence sketch the function $f(x) = \cos^{-1}(\sin x)$ in the domain $-\pi \leq x \leq \pi$

2m

1m

4m

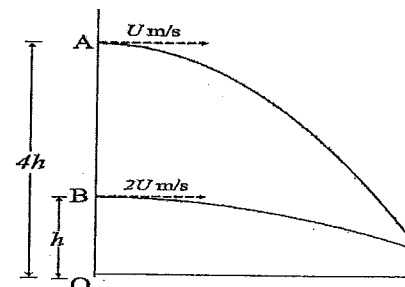
3m

2m

Question 7

Start a new page

a) Fig



A vertical building stands with its base O on the horizontal ground. A and B are two points on the building vertically above each other such that A is $4h$ metres above O and B is h metres above O.

A particle is projected horizontally from A with a speed of U m/sec and 10 seconds later a second particle is projected horizontally from B with a velocity of $2U$ m/sec. The two particles collide t seconds after the first particle is projected.

The coordinate axes are placed at O and the value of g is taken as 10 m/sec^2

(i) Show that the expressions for the horizontal and vertical displacements of each particle, t seconds after the first particle is projected are given by

$$x_A = Ut \text{ and } y_A = 4h - 5t^2 \text{ for the particle A}$$

and

$$x_B = 2U(t - 10) \text{ and } y_B = h - 5(t - 10)^2 \text{ for the particle B.} \quad 4m$$

(ii) Find the time taken for the particles to collide

1m

(iii) Find the value of h .

2m

Please turn over for 7(b)

b) (i) Show that

$$(1-x)^n \left(1 + \frac{1}{x}\right)^n = \frac{(1-x^2)^n}{x^n} \quad 1\text{m}$$

(ii) Show that

$$\frac{(1-x^2)^n}{x^n} = \sum_{r=0}^n (-1)^r \binom{n}{r} x^{2r-n} \quad 1\text{m}$$

$$(\text{note: } \binom{n}{r} = {}^nC_r)$$

(iii) By considering the coefficient of x^2 on both sides of the given identity in part (i) and using part (ii), show that

3m

$$\begin{aligned} & \binom{n}{2} \binom{n}{0} - \binom{n}{3} \binom{n}{1} + \cdots + (-1)^n \binom{n}{n} \binom{n}{n-2} \\ &= \begin{cases} 0, & \text{if } n \text{ is odd} \\ (-1)^{\frac{n+2}{2}} \binom{n}{\frac{n+2}{2}} & \text{if } n \text{ is even} \end{cases} \end{aligned}$$

END OF Paper

Qn	Solutions	Marks	Comments: Criteria
1.	$\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ $= \lim_{x \rightarrow 0} \frac{1 - (1 - 2\sin^2 x)}{x^2}$ $= \lim_{x \rightarrow 0} 2 \left(\frac{\sin x}{x} \right)^2$ $= 2 \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right)^2$ $= 2 \times 1 = 2.$	(1m)	
b)	$(e, 1)$ lies on $y = \log_e x$ for $1 = \log_e e$. also " " $xy = e$ for $e \times 1 = e$ $y = \log_e x$ $y' = \frac{1}{x}$; $y'_{at(e,1)} = \frac{1}{e}$ $y = \frac{e}{x}$ $y' = -\frac{e}{x^2}$; $y'_{at(e,1)} = -\frac{e}{e^2} = -\frac{1}{e}$ if x is the acute angle between the tangents $\tan x = \left \frac{\frac{1}{e} - (-\frac{1}{e})}{1 - \frac{1}{e^2}} \right = \left \frac{\frac{2}{e}}{e^2 - 1} \right = \frac{2e}{e^2 - 1}$ (for correct substitution)	(1m)	
c)	① $7!$ ② $5! \times 3!$ (1) (1)	(1)	

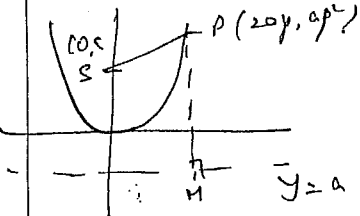
Qn	Solutions	Marks	Comments: Criteria
d)	$u = 1 - x$ $du = -dx$ $\int \frac{x dx}{\sqrt{1-x}} = \int \frac{(1-u)(-du)}{\sqrt{u}}$ $= \int (\sqrt{u} - u^{1/2}) du$ $= \frac{2}{3} u^{3/2} - 2 u^{1/2} + C$ $= \frac{2}{3} (1-x)^{3/2} - 2\sqrt{1-x} + C.$	(1m)	
2	$\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{9-2x^2}}$ $= \int_0^{\frac{3}{2}} \frac{dx}{\sqrt{2(\frac{9}{2}-x^2)}}$ $= \frac{1}{\sqrt{2}} \cdot \sin^{-1} \left(\frac{\sqrt{2}x}{3} \right) \Big _0^{\frac{3}{2}}$ $= \frac{1}{\sqrt{2}} \cdot \left(\sin^{-1} \left(\frac{\sqrt{2}}{3} \cdot \frac{3}{2} \right) - \sin^{-1} 0 \right)$ $= \frac{1}{\sqrt{2}} \left(\frac{\pi}{4} \right) = \frac{\pi}{4\sqrt{2}} = \frac{\sqrt{2}\pi}{8}$	(1m)	
b)	$P(-x) = P(x)$ (for x is a zero - given) $\therefore P(-1) = P(1) = 0$ $P(-2) = P(2) = 0$ ① The other two roots are -1 and -2 ② $P(x) = A(x-1)(x+1)(x-2)(x+2)$ $P(0) = 8$ $\therefore 8 = A(-1)(-4)$		full mark at this point

Qn	Solutions	Marks	Comments: Criteria
	$\therefore A = 2$ $P(x) = 2(x-1)(x+1)(x-2)(x+2)$ c) $\sqrt{3} \cos x - \sin x = R \cos(x+\alpha)$ $= R(\cos x \cos \alpha - \sin x \sin \alpha)$ $\therefore \sqrt{3} = R \cos \alpha \quad \therefore \tan \alpha = \frac{1}{\sqrt{3}}$ $1 = R \sin \alpha \quad \alpha = \frac{\pi}{6} \quad (1)$ $\therefore R = 2 \quad (1)$		
(ii)			end pt. 1m. turning pt 1m.
(iii)	$2 \cos(x + \frac{\pi}{6}) = 1$ $\cos(x + \frac{\pi}{6}) = \frac{1}{2}$ $x + \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{3}$ $x = 2n\pi \pm \frac{\pi}{3} - \frac{\pi}{6}$ $x = 2n\pi + \frac{\pi}{6} \text{ or } 2n\pi - \frac{\pi}{2}$		$(\frac{1}{2})$ $(\frac{1}{2})$ (1) $\leftarrow$ full marks awarded at this point.

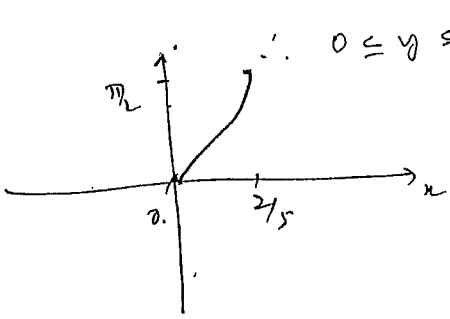
Qn	Solutions	Marks	Comments: Criteria
3	$(5+2x)^{15} = {}^{15}C_0 5^{15} + {}^{15}C_1 5^{14} (2x) + \dots$ $t_{r+1} = {}^{15}C_r 5^{15-r} (2x)^r \quad (1m)$ $t_r = {}^{15}C_{r-1} 5^{15-(r-1)} (2x)^{r-1}$ $\frac{\text{Coeff of } t_{r+1}}{\text{Coeff of } t_r} = \frac{{}^{15}C_r 5^{15-r} \cdot 2^r}{{}^{15}C_{r-1} 5^{16-r} \cdot 2^{r-1}} \quad (1m)$ $= \frac{15!}{r!(15-r)!} \cdot \frac{(r-1)!(15-(r-1))!}{15!} \cdot \frac{2}{5}$ $= \frac{16-r}{r} \cdot \frac{2}{5} = \frac{32-2r}{5r} \quad (1m)$ (i) Coeff of $t_{r+1} \geq$ Coeff of t_r when $\frac{32-2r}{5r} \geq 1$ $32-2r \geq 5r$ $7r \leq 32 \quad (4, \dots)$ i.e. when $r = 1, 2, 3, 4$ $(1m)$ Coeff of $t_{r+1} <$ Coeff of t_r for $r = 5, 6, \dots$ $\therefore t_5$ has the higher coeff. Coeff of $t_5 = {}^{15}C_4 5^{11} \cdot 2^4 \quad (1m)$		

Qn	Solutions	Marks	Comments: Criteria
11)	Vertical asymp $x = 1$ $\lim_{x \rightarrow \infty} f(x) = \infty$ $\lim_{x \rightarrow -\infty} f(x) = 0$ Horiz asymp $y = 0$ $x = 0; y = -1$ $y \neq 0$ $(2, 0)$ $y = f^{-1}(x)$ $y = f(x)$ $x = 1$ $y = 0$ b) $y = 4x - x^2$ $= x(4 - x)$ $x = 2; y = 4$ $(2, 4)$ is turning pt		

Qn	Solutions	Marks	Comments: Criteria
11)	$y = 4x - x^2$ $x = 4y - y^2$ (inv. f^{-1}) $y^2 - 4y + x = 0$ $y = \frac{4 \pm \sqrt{16 - 4x}}{2}$ $= \frac{4 \pm 2\sqrt{4 - x}}{2}$ Range of $y = f^{-1}(x)$ $= \text{Domain of } y = f(x)$ ≥ 2 $\therefore y = 2 + \sqrt{4 - x}$		
11)	$y = f(x)$ and $y = f^{-1}(x)$ meet on $y = x$ $\therefore$ their pt. of intersection is given by $4x - x^2 = x$ $3x - x^2 = 0$ $x = 0; x = 3$ $x = 0$ is not in the domain. $\therefore (3, 3)$ is the pt. of intersection		

Qn	Solutions	Marks	Comments: Criteria
5.	$x^2 = 4ay$ Eqn of the tangent at $P(2ap, ap^2)$ $y = px - ap^2$ — (1) Eqn of the tangent at Q is $y = qx - aq^2$ — (2) (1) & (2) meet at $px - ap^2 = qx - aq^2$ $(p-q)x = a(p^2 - q^2)$ $x = a(p+q) \quad (p \neq q)$ (ii) Sub in (1) $y = pa(p+q) - ap^2$ $= apq$  $SP = PM$ by def. $= ap^2 + a$ $= a(p^2 + 1)$ (iii) $SP + SQ = 4a$ $a(p^2 + 1) + a(q^2 + 1) = 4a$ $\therefore p^2 + q^2 = 2$ — (1m) $\therefore$ T moves so that $x = a(p+q)$ and $y = apq$ and $p^2 + q^2 = 2$ $(p+q)^2 = p^2 + q^2 + 2pq$ $(\frac{x}{a})^2 = 2 + 2apq$ is known attempt to compare 1m		

Qn	Solutions	Marks	Comments: Criteria
b)	Let $x = a \cos(\omega t + C)$ $\frac{x}{a} = \cos(\omega t + C)$ at $t=0$; $x=1$ $\therefore 1 = a \cos C$ — (1) $x' = -a\omega \sin(\omega t + C)$ $\frac{x'}{a\omega} = -\sin(\omega t + C)$ at $t=0$; $x'=1$ $\therefore 1 = -a\omega \sin C$ — (2) $\therefore \tan C = -1$ $C = -\pi/4$ $\therefore x = \sqrt{2} \cos(t - \pi/4)$ c) $f(x) = x^3 + ax^2 + bx + c$ $f'(a) = 0$; $f'(b) = 0$ $\therefore a$ & b are roots of $f'(x) = 3x^2 + 2ax + b = 0$ $\therefore a + b = -\frac{2a}{3}$ $f''(x) = 6x + 2a$ possible pt. of inflexion is when $f''(x) = 0$ $6x + 2a = 0$ $x = -\frac{2a}{6}$ $= -\frac{a}{3}$ $= \frac{a+b}{2}$	2m 1m	

Qn	Solutions	Marks	Comments: Criteria
Q.6	$\frac{\alpha+\beta}{2}$ is between α & β $f''(\alpha) < 0$; $f''(\frac{\alpha+\beta}{2}) = 0$; $f''(\beta) > 0$ $\therefore x = \frac{\alpha+\beta}{2}$ is a pt. of inflexion (1m) $f(x) = \frac{1}{2} \cos^{-1}(1-3x)$ Dom: $-1 \leq 1-3x \leq 1$ $-2 \leq -3x \leq 0$ $0 \leq x \leq \frac{2}{3}$ Range: when $x=0$; $f(0) = \frac{1}{2} \cos^{-1}(1)$ $= 0$ when $x = \frac{2}{3}$; $f(\frac{2}{3}) = \frac{1}{2} \cos^{-1}(1-2)$ $= \frac{\pi}{2}$ $\therefore 0 \leq y \leq \frac{\pi}{2}$ 		

Qn	Solutions	Marks	Comments: Criteria
Q.7	$\int_0^3 \frac{x^2 dx}{\sqrt{9-x^2}}$ Let $x = 3 \sin \theta$ $dx = 3 \cos \theta d\theta$ $x=0$; $\theta=0$ $x=3$; $\theta = \pi/2$ $\therefore \int_0^3 \frac{x^2 dx}{\sqrt{9-x^2}}$ $= \int_0^{\pi/2} \frac{9 \sin^2 \theta \cdot 3 \cos \theta d\theta}{\sqrt{9-9 \sin^2 \theta}}$ (1m) $= 9 \int_0^{\pi/2} \sin^2 \theta d\theta$ (1/2 m) $= 9 \int_0^{\pi/2} \frac{(1 - \cos 2\theta)}{2} d\theta$ (1m) $= \frac{9}{2} \left(\theta - \frac{\sin 2\theta}{2} \right)_0^{\pi/2}$ (1m) $= \frac{9}{2} \cdot \frac{\pi}{2} = \frac{9\pi}{4}$ (1/2 m)		

Qn	Solutions	Marks	Comments: Criteria
c)	$f(x) = \cos^{-1}(\sin x)$ $f'(x) = \frac{-1}{\sqrt{1-\sin^2 x}} \cdot \cos x$ $= \frac{\cos x}{ \cos x } \quad (1 \frac{1}{2} m)$ $= -1 \quad \text{when } \cos x > 0 \quad (\frac{1}{2} m)$ $= 1 \quad \text{when } \cos x < 0$ $f'(x) = -1; \text{ when } -\frac{\pi}{2} < x < \frac{\pi}{2}$ $f'(x) = 1 \quad \text{when } \frac{\pi}{2} < x < \pi \text{ or } -\pi < x < -\frac{\pi}{2} \quad \left. \vphantom{\frac{\pi}{2} < x < \pi} \right\} 1 m$ $f(\frac{\pi}{2}) = \pi; f(0) = \frac{\pi}{2}; f(\frac{3\pi}{2}) = 0$ $f(-\pi) = \frac{\pi}{2}; f(\pi) = \frac{\pi}{2}$ 		

Qn	Solutions	Marks	Comments: Criteria
7	A $\dot{x} = 0$ $\ddot{x} = \cos t$ $\text{at } t=0; x=4$ $\therefore x = u$ $x = ut + c$ $t=0; x=0$ $\therefore x = ut$ $\dot{y} = -10$ $y = -10t + c$ $t=0; y=0 \therefore c=0$ $y = -10t$ $y = -5t^2 + c \quad \begin{cases} t=0 \\ y=4h \end{cases}$ $\therefore c = 4h$ $\therefore y = 4h - 5t^2$ B $\dot{x} = 0$ $\ddot{x} = \cos t$ $t=10; x=2u$ $\therefore x = 2u$ $x = 2ut + c$ $\text{at } t=10; x=0$ $0 = 20u + c$ $\therefore x = 2ut - 20u$ $= 2u(t-10)$ $\dot{y} = -10$ $y = -10t + c$ $t=10; y=0$ $\therefore c = 100$ $y = -10t + 100$ $y = -5t^2 + 100t + c$ $t=10; y=h$ $\therefore h = -500 + 1000 + c$ $\therefore c = h - 500$ $\therefore y = -5t^2 + 100t + h - 500$ $= h - 5(t^2 - 20t + 100)$ $= h - 5(t-10)^2$		

Qn	Solutions	Marks	Comments: Criteria
	where they collide $x_A = x_B ; \quad ut = 2ut - 20u$ $\therefore t = 20 \text{ sec}$ $y_A = y_B ; \quad 4h - 5t^2 = h - 5(t-10)^2$ $3h = 5(t^2 - (t-10)^2)$ $t = 20$ $3h = 1500$ $h = 500 \text{ m.}$ b) $(1-x)^n \left(1+\frac{1}{x}\right)^n$ $= (1-x)^n \frac{(x+1)^n}{x^n}$ $= \frac{(1-x^2)^n}{x^n}$ ① $\frac{(1-x^2)^n}{x^n} = \frac{1}{x^n} \left({}^nC_0 - {}^nC_1 x^2 + {}^nC_2 x^4 - \dots \right)$ $= \frac{1}{x^n} \sum_{r=0}^n {}^nC_r (-1)^r x^{2r}$ $= \sum_{r=0}^n (-1)^r {}^nC_r x^{2r-n}$		

Qn	Solutions	Marks	Comments: Criteria
	Coeff of x^2 is $\sum_{r=0}^n (-1)^r {}^nC_r x^{2r-n} \text{ is when}$ $2r-n = 2 \quad \text{or when } r = \frac{n+2}{2}$ $\therefore \text{Coeff is } (-1)^{\frac{n+2}{2}} {}^nC_{\frac{n+2}{2}}$ This is zero when n is odd; for $\frac{n+2}{2}$ will not be a whole number $\therefore (-1)^{\frac{n+2}{2}} \cdot {}^nC_{\frac{n+2}{2}}$ is the Coeff of x^2 when n is even. L.H.S. $\left({}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 - \dots \right) \left({}^nC_0 + {}^nC_1 \frac{1}{x} + {}^nC_2 \frac{1}{x^2} + \dots \right)$ Coeff of x^2: $\left({}^nC_2 \cdot {}^nC_0 \right) - \left({}^nC_3 \cdot {}^nC_1 \right) + \left({}^nC_4 \cdot {}^nC_2 \right) - \dots + (-1)^n \cdot {}^nC_n \cdot {}^nC_{n-2}$ Hence the result.		